

FP-AIM: Floorplan-Aware Alignment and Implicit Mapping for Sparse-Ranging Micro-Quadrotors

Anonymous Submission
Paper under double-anonymous review

Abstract—Micro-quadrotors are well suited for confined indoor environments, but their strict payload constraints often limit onboard perception to sparse range sensing and lightweight onboard odometry. Sparse range returns provide limited geometric evidence, making room-scale structural consistency difficult to maintain from onboard measurements alone. Architectural floorplans provide complementary global structure, but using them for mapping requires aligning the local odometry frame to the floorplan and handling discrepancies between planned structure and observed geometry. We present FP-AIM, an integrated floorplan-aware system that uses a 3D floorplan as a structural reference for sparse-range local-to-world registration and floorplan-aware implicit SDF mapping. The alignment stage estimates the local-to-world transform using a particle filter over a 2D floorplan SDF, combining sparse-beam likelihoods with candidate verification and recovery to reduce unsupported convergence and persistent tracking failure. The mapping stage warm-starts an SDF field from a floorplan-derived 3D SDF and updates it online with conflict-aware supervision, balancing floorplan-consistent structure with measurement-supported local deviations. Experiments on simulation-generated indoor sequences demonstrate improved alignment reliability and reconstruction quality, and a qualitative real-world deployment shows coherent reconstruction under sparse range sensing.

Index Terms—Micro-quadrotor, Sparse Range Sensing, Floorplan, Global Alignment, Implicit Mapping

I. INTRODUCTION

Micro-quadrotors have emerged as a promising platform for autonomous operation in narrow and cluttered indoor environments [1]–[3], where agile navigation and a small form factor are essential. Unlike Unmanned Ground Vehicles (UGVs), which are limited by ground obstacles such as stairs, and conventional Unmanned Aerial Vehicles (UAVs), which are often impractical in confined spaces due to their size and strong downwash, micro-quadrotors can operate effectively in tight indoor settings. In such environments, complex structures and frequent occlusions make reliable 3D mapping essential. However, micro-quadrotors are subject to strict size, weight, and power (SWaP) constraints, which severely limit sensing capability [1], [3]. As a result, the system must rely on extremely sparse range observations from lightweight sensors, and such sensing sparsity fundamentally challenges accurate and consistent indoor 3D mapping. Under extremely sparse range observations, the choice of map representation becomes critical for reliable indoor mapping. Explicit representations such as occupancy grids and TSDF/ESDF voxel maps are widely used in robotics [4]–[7], but they often become fragmented or incomplete in sparsely observed regions. Implicit representations, particularly signed distance functions (SDFs),

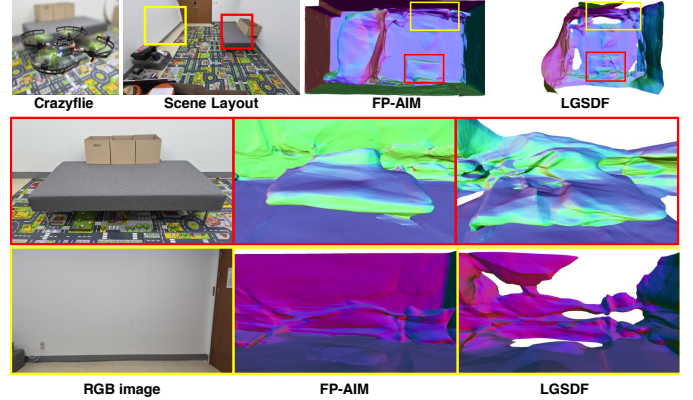


Fig. 1. Qualitative results of the full-pipeline deployment in the real-world M09 scene. Using a Crazyflie equipped with Flow-deck local odometry and sparse Multi-ranger measurements, the system shows coherent room-scale reconstruction.

provide a continuous formulation for representing and querying geometry beyond discrete measurements [8]–[10]. However, under severe sparsity, implicit models alone struggle to maintain global consistency and can become unstable during optimization. This suggests that representation alone is insufficient in such settings, motivating the incorporation of additional structural priors. In this work, we leverage architectural floorplans as a source of global structural information to enable more consistent reconstruction under extreme sensing sparsity [11]–[14].

While floorplans provide valuable global structural information for indoor environments [11]–[14], effectively leveraging them under extreme sensing sparsity introduces two key challenges. First, the floorplan defines a coordinate frame independent of the robot’s local odometry frame, so the system must estimate an accurate global alignment before the mapping module can use the floorplan. Under sparse and partially occluded range observations, this alignment becomes highly ambiguous due to limited geometric constraints. Second, discrepancies between the floorplan and real-world observations are inevitable, as floorplans typically encode static structural layouts and may differ from the currently observed environment [11]. For example, a cabinet or newly placed obstacle may produce a short range return in a region marked as free space in the floorplan, leading to conflicting geometric constraints that can degrade reconstruction quality if not properly handled. These challenges highlight the need

for a framework that can jointly handle uncertain global alignment and floorplan–observation inconsistencies under extreme sensing sparsity.

To address these challenges, we present **FP-*AIM***, a floorplan-aware framework for micro-quadrotors that jointly performs global alignment and implicit SDF mapping under sparse range observations. **FP-*AIM*** uses the floorplan as a shared structural reference across both stages, enabling reliable alignment while guiding consistent reconstruction. In the first stage, we estimate the alignment between the local odometry frame and the floorplan-aligned world frame using a particle filter over a 2D floorplan SDF, combining sparse-beam likelihoods with candidate verification and recovery to reduce unsupported convergence under sparse and partially inconsistent measurements. In the second stage, we perform floorplan-aware implicit mapping through conflict-aware online refinement, allowing the model to preserve floorplan-derived global structure while adapting to newly observed local geometry. Together, these components enable more stable global alignment and consistent 3D reconstruction under extreme sensing sparsity.

The main contributions of this work are as follows:

- We present **FP-*AIM***, a floorplan-aware integrated system for sparse-ranging micro-quadrotors that uses a 3D floorplan as a shared structural reference for both global alignment and implicit SDF mapping.
- We present a floorplan-aware global alignment module that integrates sparse range likelihoods, odometry-driven particle propagation, candidate verification, and recovery into a practical lifecycle for initializing downstream floorplan-aware mapping.
- We present a floorplan-aware implicit mapping strategy that warm-starts the implicit field from a floorplan-derived structural SDF and refines it online using sparse measurements with conflict-aware hybrid supervision.
- We evaluate global alignment and mapping on simulation generated datasets and qualitatively demonstrate the full pipeline on a real Crazyflie platform under sparse onboard range sensing.

II. RELATED WORK

Mapping under Sparse Sensing. Micro-quadrotors operate under stringent size, weight, and power constraints, which motivates lightweight perception and mapping from sparse range or low-power depth sensing [1]–[3]. Sparse-ranging SLAM methods exploit structural regularity, such as soft Manhattan-world constraints [15] or line structures with a four-point LiDAR [16], to compensate for limited measurements. These systems are efficient, but they primarily maintain explicit or discrete representations that remain tied to directly observed regions and map resolution. In contrast, **FP-*AIM*** targets online implicit SDF reconstruction from sparse range observations, using a floorplan-derived structural reference to stabilize both global alignment and mapping.

Floorplan Localization and Alignment. Architectural floorplans provide compact structural references for indoor lo-

calization. Prior methods estimate poses in a floorplan frame using semantic sensor fusion [12], RGB-to-floorplan depth matching and temporal filtering [13], semantic-ray evidence [14], or floorplan-prior map construction [17]. These methods show the value of floorplans for localization, but floorplan-aware mapping also requires a stable local-to-world transform that registers the robot’s local odometry frame to the floorplan-aligned world frame. Plan-aware SLAM addresses related plan-to-observation alignment issues [11], but it targets richer SLAM settings. In contrast, **FP-*AIM*** estimates this transform from sparse range observations and local odometry increments, and reuses it as the structural coordinate frame for floorplan-aware implicit SDF mapping.

Online Implicit SDF Mapping. Implicit SDF maps provide compact and continuous distance queries for reconstruction and planning. *iSDF* learns an online neural SDF from posed depth images [8], *HIO-SDF* combines hierarchical voxel structures with local neural detail [9], *LGSDf* improves continual ESDF learning using locally updated anchors [10], and *AiSDF* incorporates structural assumptions into neural SDF reconstruction [18]. These methods demonstrate the potential of online implicit mapping, but they generally assume richer geometric input, such as dense depth, RGB-D streams, or observation-rich local updates. They do not explicitly handle conflicts between architectural floorplans and sparse online range observations. **FP-*AIM*** addresses this setting by coupling floorplan-aware global alignment with conflict-aware implicit SDF refinement.

III. METHOD

We propose **FP-*AIM***, a two-stage floorplan-aware framework for sparse-range micro-quadrotor mapping. Given local odometry and sparse range observations from a micro-quadrotor, **FP-*AIM*** leverages a 3D floorplan as a structural reference to establish a globally consistent reference frame and reconstruct an implicit SDF map. The floorplan serves as a shared structural reference across the two stages. The first stage estimates the local-to-world alignment by searching over the unknown horizontal translation and yaw, using sparse horizontal range evidence and a horizontal floorplan SDF slice to score and validate candidate transforms. The second stage uses the alignment transform to warm-start an implicit SDF from a floorplan-derived 3D SDF and refine it online with sparse measurements, balancing floorplan-derived structure with measurement-supported deviations. The overall pipeline is shown in Fig. 2.

A. Floorplan-Aware Global Alignment

The global alignment stage aims to estimate a rigid transform that aligns the local frame of the micro-quadrotor with the floorplan-aligned world frame. Concretely, given local body poses from the micro-quadrotor, the alignment module estimates a local-to-world transform such that

$${}^W\mathbf{T}_B(t) = {}^W\mathbf{T}_L^L \mathbf{T}_B(t), \quad (1)$$

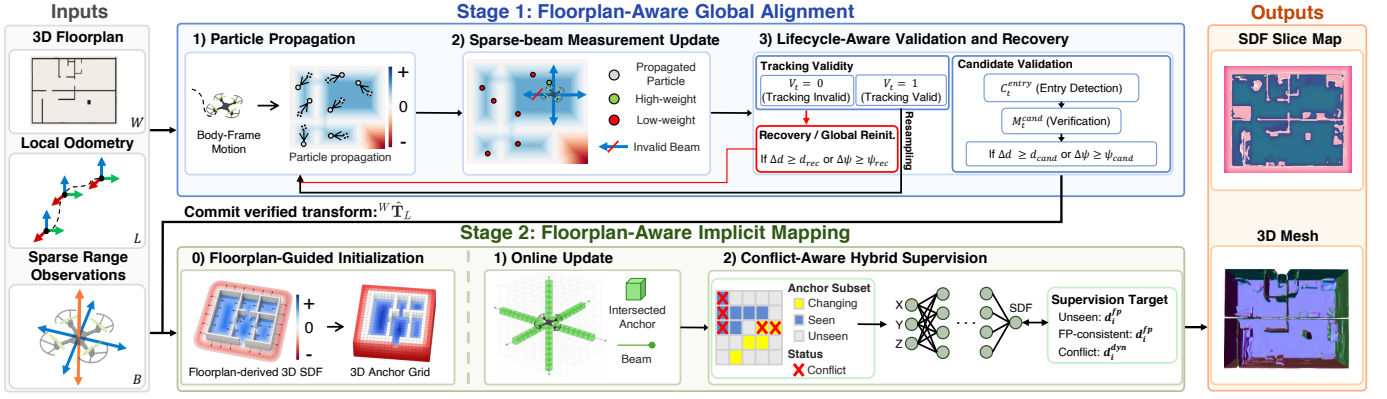


Fig. 2. Overview of FP-AIM. The first stage estimates the local-to-world alignment using sparse range observations, local odometry increments, and a 2D floorplan SDF. The second stage warm-starts an implicit SDF from the floorplan-derived structural field and refines it online with sparse measurements and conflict-aware supervision.

where ${}^X\mathbf{T}_Y \in SE(3)$ denotes the rigid transform from frame Y to frame X , and the frame symbols L , W , and B denote the local odometry frame, the floorplan-aligned world frame, and the body frame, respectively. Here, ${}^W\mathbf{T}_L$ is the unknown alignment transform to be estimated, and ${}^L\mathbf{T}_B(t)$ and ${}^W\mathbf{T}_B(t)$ denote the body pose at time t expressed in L and W , respectively. The local odometry stream provides ${}^L\mathbf{T}_B(t)$, whereas the corresponding world-frame body pose ${}^W\mathbf{T}_B(t)$ is not directly observed. We thus estimate body pose hypotheses in the floorplan-aligned world frame and recover the local-to-world transform from a verified hypothesis and the corresponding local odometry pose.

To make this inference tractable under sparse range sensing, we assume gravity-aligned local and world frames. The floorplan-aligned world frame defines the vertical z axis, while the local odometry frame shares the same vertical direction but remains offset by an unknown horizontal translation and yaw. Under this assumption, estimating the world-frame body pose during alignment reduces to estimating its horizontal translation and yaw while fixing the sensing height. We therefore search over a planar 3-DoF state to construct ${}^W\hat{\mathbf{T}}_B(t)$.

Even in this reduced search space, sparse range observations often provide insufficient information to uniquely determine the world-frame body pose. Moreover, the observed ranges may include returns from scene structures and objects that are not represented in the floorplan, introducing additional uncertainty into the alignment process. To maintain multiple hypotheses of ${}^W\hat{\mathbf{T}}_B(t)$ under these uncertainties, we adopt a standard Monte Carlo localization (MCL) formulation [19], a particle-filter-based method for robot localization. MCL represents the posterior belief over possible body poses using a set of particles. Following this representation, we denote the particle set at time t by

$$S_t = \{\mathbf{q}_t^i\}_{i=1}^N, \quad (2)$$

where N is the number of particles and $\mathbf{q}_t^i$ is the planar state of the i -th hypothesis for ${}^W\hat{\mathbf{T}}_B(t)$. Each particle state is defined

as

$$\mathbf{q}_t^i = [\mathbf{p}_t^i, \psi_t^i]^\top, \quad \mathbf{p}_t^i = [x_t^i, y_t^i]^\top, \quad (3)$$

where $\mathbf{p}_t^i$ and ψ_t^i denote the horizontal position and yaw of the i -th particle in W at time t , respectively.

At each time step, MCL updates the particle set through propagation, weight update, and resampling,

$$S_{t-1} \rightarrow S'_t \rightarrow \{(\bar{\mathbf{q}}_t^i, w_t^i)\}_{i=1}^N \rightarrow S_t. \quad (4)$$

The propagation applies the motion model to the previous particle set S_{t-1} and produces the propagated set $S'_t = \{\bar{\mathbf{q}}_t^i\}_{i=1}^N$. The weight update evaluates the propagated particles with the observation model and assigns likelihood weights w_t^i . Resampling then draws the next particle set S_t from this weighted set, retaining high-weight body pose hypotheses more often while making low-weight hypotheses less likely to remain. Thus, strong observations concentrate the particles around one hypothesis of ${}^W\hat{\mathbf{T}}_B(t)$, whereas ambiguous observations allow several such hypotheses to persist. The following subsections instantiate this MCL flow for floorplan-aware global alignment by specifying the propagation input, the sparse-beam floorplan observation model, and the additional validation and recovery checks.

Particle propagation. The motion model propagates each particle using a planar body-frame motion increment computed from consecutive local body poses and expressed in the forward-left-up convention,

$$\mathbf{u}_t = [\Delta f_t \quad \Delta l_t \quad \Delta \psi_t]^\top, \quad (5)$$

where Δf_t and Δl_t denote frame-to-frame forward and left translation increments in the previous body frame, respectively, and $\Delta \psi_t$ denotes the yaw increment about the gravity-aligned vertical axis. For each particle $\mathbf{q}_{t-1}^i$, we propagate the planar state as

$$\bar{\mathbf{p}}_t^i = \mathbf{p}_{t-1}^i + \mathbf{R}(\psi_{t-1}^i) \left(\begin{bmatrix} \Delta f_t \\ \Delta l_t \end{bmatrix} + \epsilon_{p,t}^i \right), \quad (6)$$

$$\bar{\psi}_t^i = \psi_{t-1}^i + \Delta \psi_t + \epsilon_{\psi,t}^i, \quad (7)$$

where $\bar{\mathbf{p}}_t^i$ and $\bar{\psi}_t^i$ denote the propagated position and yaw before the range-measurement update, $\mathbf{R}(\psi)$ is the 2D rotation matrix for yaw angle ψ , $\epsilon_{p,t}^i \sim \mathcal{N}(\mathbf{0}, \Sigma_p)$ is body-frame translational process noise, and $\epsilon_{\psi,t}^i \sim \mathcal{N}(0, \sigma_\psi^2)$ is yaw process noise.

Sparse-beam measurement update. The sparse-beam measurement update evaluates each propagated particle by comparing the current horizontal range observations with ranges predicted from the horizontal floorplan SDF slice. For simplicity, we omit the time index on beam-level range quantities. For each horizontal beam k , the range sensor measures r_k along a known beam direction in B . To predict the corresponding range for $\bar{\mathbf{q}}_t^i$, we use the horizontal floorplan SDF slice extracted at the sensing height. We denote this slice by $S_{\text{FP}} : \mathbb{R}^2 \rightarrow \mathbb{R}$, where $S_{\text{FP}}(\mathbf{x})$ returns the signed distance to the nearest floorplan wall at a horizontal query location $\mathbf{x}$. We compute the predicted range as

$$\hat{r}_k^i = h_k(\bar{\mathbf{q}}_t^i, S_{\text{FP}}), \quad (8)$$

where $h_k(\bar{\mathbf{q}}_t^i, S_{\text{FP}})$ denotes SDF-based raycasting along beam k from $\bar{\mathbf{q}}_t^i$ on S_{FP} . We compute an unnormalized particle weight using a Gaussian residual likelihood over valid closed-return beams,

$$\tilde{w}_t^i = \chi_t^i \exp\left(-\sum_{k \in \mathcal{K}_t} \frac{(r_k - \hat{r}_k^i)^2}{2\sigma_{\text{hit}}^2}\right), \quad (9)$$

where $\mathcal{K}_t$ denotes the set of valid closed-return beams within the sensing horizon at time t , excluding no-return and maximum-range beams; σ_{hit} controls the residual tolerance of the beam model; and $\chi_t^i \in \{0, 1\}$ suppresses particles outside the floorplan-valid support. We then normalize the particle weights, $w_t^i \propto \tilde{w}_t^i$ with $\sum_{i=1}^N w_t^i = 1$, to obtain the frame-wise belief used for resampling and lifecycle validation.

Lifecycle-aware validation and recovery. The sparse-beam measurement update provides frame-wise evidence for each propagated particle, but posterior concentration at a single time step does not by itself justify confirming a global alignment. Under sparse horizontal beams, locally consistent floorplan matches can transiently concentrate the particle belief around an incorrect world-frame body pose. We therefore verify a candidate pose estimate over additional motion and reinitialize the particle set only when the current belief remains persistently unsupported.

To verify a candidate pose before alignment-transform estimation, we evaluate three validation criteria from the current weighted particle belief: beam support, particle-set cohesion, and posterior uniqueness. Beam support checks whether the valid closed-return ranges agree with the ranges predicted by the maximum-weight particle. Particle-set cohesion checks whether high-weight particles are concentrated in horizontal position and yaw, while posterior uniqueness checks whether the weighted belief contains a single dominant mode rather than multiple spatially separated high-weight clusters. We apply strict criteria for candidate selection and looser criteria for candidate verification and tracking-validity monitoring.

When the strict criteria are satisfied, we select the maximum-weight particle as a candidate pose and retain its pose correspondence,

$$\mathcal{P}_c = \left({}^W\hat{\mathbf{T}}_B(t_c), {}^L\mathbf{T}_B(t_c) \right), \quad (10)$$

where t_c denotes the candidate selection time. The candidate is then verified over subsequent motion. If the looser criteria remain satisfied until the accumulated translation or yaw change exceeds d_{cand} or ψ_{cand} , respectively, we confirm the candidate for alignment-transform estimation. If support is lost before this motion requirement is met, we discard the candidate and continue filtering.

For recovery, the tracking-validity criterion V_t distinguishes supported tracking from persistent tracking failure. When $V_t = 1$, we treat the current belief as valid and resample particles according to their normalized weights. When $V_t = 0$, we accumulate unsupported translation and yaw change. If the accumulated unsupported motion exceeds d_{rec} or ψ_{rec} , respectively, we globally reinitialize the particles over floorplan-valid free space.

Alignment transform confirmation. Once candidate verification succeeds, we compute the confirmed local-to-world transform from the pose correspondence $\mathcal{P}_c$ rather than from the final verification frame. This choice prevents small changes in the particle estimate during verification from perturbing the final alignment. We construct ${}^W\hat{\mathbf{T}}_B(t_c)$ from the candidate planar state using its horizontal position and yaw, the fixed sensing height, and the roll and pitch from the local odometry estimate.

$${}^W\hat{\mathbf{T}}_L = {}^W\hat{\mathbf{T}}_B(t_c) \left({}^L\mathbf{T}_B(t_c) \right)^{-1}. \quad (11)$$

The resulting transform converts subsequent local odometry poses into the floorplan-aligned world frame for floorplan-aware implicit mapping.

B. Floorplan-Aware Implicit Mapping

To reconstruct an online implicit SDF map in the floorplan-aligned world frame, we transform local odometry poses using the confirmed local-to-world transform ${}^W\hat{\mathbf{T}}_L$ and process sparse range observations in the aligned frame. The mapping stage must reconcile two sources of geometric supervision: a floorplan-derived structural reference and local geometry accumulated from sparse measurements. These sources can disagree in regions containing unmodeled objects or layout changes. The mapping objective is to preserve room-scale structural consistency while allowing measurement-supported local deviations.

Following the anchor-based continual implicit mapping framework of LGSDF [10], we adopt its single-MLP and anchor-grid structure to maintain local geometric supervision while representing a continuous scene-wide SDF. The MLP $\Phi_\theta : \mathbb{R}^3 \rightarrow \mathbb{R}$ maps a 3D query point to its signed distance value and represents the final continuous map. The regular anchor grid $\mathcal{G} = \{\mathbf{p}_i\}_{i=1}^{N_a}$ stores locally accumulated supervision, where $\mathbf{p}_i$ denotes an anchor location and N_a denotes the number of anchors. Whereas LGSDF updates anchors from

posed dense depth observations, our setting uses sparse range beams and a floorplan-derived structural field. We therefore maintain separate floorplan-derived and measurement-updated anchor states so that training targets can be selected according to floorplan–observation consistency.

For each anchor, we maintain a measurement-updated state, a floorplan-derived SDF value, and an observation flag,

$$\mathbf{s}_i^{\text{dyn}} = (d_i^{\text{dyn}}, w_i^{\text{dyn}}, \mathbf{g}_i^{\text{dyn}}), \quad d_i^{\text{fp}}, \quad m_i^{\text{seen}} \in \{0, 1\}, \quad (12)$$

where the superscripts *dyn* and *fp* denote measurement-updated and floorplan-derived quantities, respectively; the entries *d*, *w*, and *g* denote an SDF value, an observation weight, and an SDF gradient; and m_i^{seen} indicates whether online measurements have updated anchor *i*. This separation allows us to use floorplan-derived targets for anchors without online measurement support and measurement-updated targets where sparse observations indicate local deviations from the floorplan.

Floorplan-guided initialization. To provide a coherent structural starting field before measurement-driven mapping updates begin, we initialize the anchor grid and the implicit field from a precomputed floorplan-derived 3D SDF. We load the precomputed SDF values into the anchor grid as d_i^{fp} . We then warm-start the MLP by regressing this structural field over sampled anchors:

$$\mathcal{L}_{\text{warm}} = \frac{1}{|\mathcal{B}_{\text{warm}}|} \sum_{i \in \mathcal{B}_{\text{warm}}} |\Phi_{\theta}(\mathbf{p}_i) - d_i^{\text{fp}}|, \quad (13)$$

where $\mathcal{B}_{\text{warm}}$ denotes the sampled warm-start anchor set. The warm start initializes a continuous floorplan-aligned SDF from the loaded structural field.

Measurement-driven online update. We use current sparse range observations to update anchors intersected by valid closed-return beams. We select anchor candidates by traversing the anchor grid along each valid beam with the Amanatides–Woo voxel traversal algorithm [20]. For each selected anchor, we compute a measurement-derived signed-distance target from the closest valid beam endpoint. Anchors between the sensor origin and the endpoint receive positive targets, and anchors behind the endpoint within the truncation band receive negative targets. We update $\mathbf{s}_i^{\text{dyn}}$ with this target and its measurement-derived gradient following the local update rule of LGSDF [10], and set $m_i^{\text{seen}} = 1$. This update records measurement-supported local geometry while leaving the floorplan-derived value d_i^{fp} unchanged as a structural reference.

Conflict-aware hybrid supervision. We convert the stored anchor states into SDF training targets, using floorplan-derived supervision for anchors without online measurement support and measurement-updated supervision where sparse observations support local deviations from the floorplan.

At each mapping step, we form an online training batch $\mathcal{B}_t$ from three anchor subsets: changing anchors $\mathcal{T}_{\text{chg}}$, seen anchors $\mathcal{T}_{\text{seen}}$, and floorplan-guided unseen anchors $\mathcal{T}_{\text{unseen}}$.

The changing subset prioritizes anchors whose measurement-updated SDF values change in the current frame. The seen subset samples anchors that have been updated by online measurements at least once. The unseen subset samples anchors without online measurement support from floorplan-structured regions, including anchors near floorplan-derived surfaces and anchors with negative floorplan-derived SDF values.

For anchors updated by online measurements, we identify measurement-supported conflicts with the floorplan-derived field as

$$m_i^{\text{conf}} = \mathbb{I} \left[w_i^{\text{dyn}} \geq \tau_w \wedge d_i^{\text{fp}} - d_i^{\text{dyn}} > \tau_c \right], \quad (14)$$

where $\mathbb{I}[\cdot]$ denotes the indicator function, τ_w denotes the minimum accumulated measurement-weight threshold, and τ_c denotes the conflict margin. This mask marks observed anchors where accumulated sparse observations indicate a floorplan–measurement mismatch, with the measurement-updated SDF placing the local surface closer than the floorplan-derived field by more than τ_c .

We select the SDF target for each anchor as

$$\bar{d}_i = \begin{cases} d_i^{\text{fp}}, & i \in \mathcal{T}_{\text{unseen}}, \\ d_i^{\text{fp}}, & m_i^{\text{conf}} = 0 \text{ and } |d_i^{\text{fp}}| \leq \tau_s, \\ d_i^{\text{dyn}}, & \text{otherwise,} \end{cases} \quad (15)$$

where τ_s denotes the floorplan near-surface threshold. This rule uses floorplan-derived targets for anchors without online measurement support and for floorplan-consistent anchors near floorplan-derived surfaces. It uses measurement-updated targets for the remaining anchors with $m_i^{\text{seen}} = 1$, including anchors with measurement-supported floorplan mismatches.

We optimize the implicit field with the selected targets:

$$\mathcal{L}_{\text{map}} = \frac{1}{|\mathcal{B}_t|} \sum_{i \in \mathcal{B}_t} \rho_i \ell_{\text{sdf}}(\Phi_{\theta}(\mathbf{p}_i), \bar{d}_i) + \lambda_g \mathcal{L}_{\text{grad}}^{\text{seen}} + \lambda_e \mathcal{L}_{\text{eik}}^{\text{seen}}, \quad (16)$$

where ρ_i denotes the anchor-wise supervision weight, ℓ_{sdf} denotes the SDF regression loss, $\mathcal{L}_{\text{grad}}^{\text{seen}}$ and $\mathcal{L}_{\text{eik}}^{\text{seen}}$ denote the gradient and Eikonal regularization terms evaluated on seen anchors, and λ_g and λ_e denote their weights. We apply these regularization terms only to anchors with $m_i^{\text{seen}} = 1$, where measurement-updated gradients provide local differential constraints. The optimized MLP provides the continuous SDF map; we query it on a regular grid to obtain an ESDF volume and extract the zero-level set as a mesh using Marching Cubes [21].

IV. EXPERIMENTS

A. Experimental Setup

Datasets. We use three OmniGibson [22] indoor scenes spanning a spectrum of scene scale and object density:

- Rs_int:** the smallest scene but with very densely placed objects, yielding frequent floorplan–observation conflicts and abundant returns from unmodeled clutter.
- Merom_0_int:** intermediate in both extent and object density, with unmodeled objects that induce repeated tracking-invalid intervals at low sensing heights.

- c) *Ihlen_0_int*: the largest scene with large room units but sparse, small objects, so few beams return from unmodeled geometry and floorplan–observation conflicts carry low measurement confidence.

For each scene we construct the floorplan-derived structural fields used by the two stages: a horizontal 2D SDF slice for global alignment and a 3D SDF for floorplan-aware mapping. Sparse range observations and ground-truth poses are generated in NVIDIA Isaac Sim and replayed for all methods on an offboard server, providing identical input sequences and separating sequence generation from algorithm execution.

Evaluation Protocol. We use three stage-specific evaluation sets:

- a) *Global-alignment set*: three sensing heights ($h_{030} / h_{050} / h_{100} \approx 0.3 / 0.5 / 1.0$ m) per scene, with 20 episodes each (5 validation / 15 test), reported over seeds 42–44, yielding 405 test runs per method; parameters and lifecycle thresholds are frozen on the validation split.
- b) *Mapping-only set*: one six-beam trajectory per scene with world-aligned ground-truth poses, isolating reconstruction quality from alignment error.
- c) *Integrated set*: one trajectory per scene linking a verified-convergence Stage 1 segment to the mapping trajectory.

For global alignment and the Stage 1 segment of the integrated set, we use a drift-free synthetic local-odometry stream (a fixed unknown rigid transform applied to the ground-truth trajectory), isolating the local-to-world alignment problem from long-term odometry drift.

Baselines. We compare against two particle-filter localizers that share the 2D floorplan SDF, odometry, sparse range observations, particle count, sensing horizon, and base convergence check for evaluating global alignment:

- *MCL* [19]: the classical Monte Carlo localization formulation, weighting particles by sparse-beam likelihood and declaring convergence directly on the base convergence check.
- *AMCL* [23]: an adaptive variant using KLD-sampling that adjusts the particle-set size online, improving reliability over MCL at the cost of slower convergence.

Both declare convergence directly from the base check, whereas FP-AIM requires candidate verification over accumulated motion; together these span classical and adaptive particle-filter localization.

We evaluate the mapping performance of the proposed method against three mappers spanning explicit and implicit paradigms, all using the pose source of each evaluation set and the same sparse six-beam observations without dense depth:

- *nvblox* [7]: a GPU-accelerated explicit volumetric ESDF mapper that accumulates distance fields near directly observed surfaces.
- *iSDF* [8]: an online neural SDF learning a continuous field from posed range, providing smooth interpolation.
- *LGSDf* [10]: a continual neural ESDF that stabilizes learning through local anchor updates but does not reconcile

TABLE I
HYPERPARAMETERS, FROZEN ON THE VALIDATION SPLIT AND FIXED ACROSS SCENES.

Symbol	Meaning	Value
<i>Particle filter</i>		
N	particles	10,000
$z_{\max}$	sensing horizon	4.0 m
σ_{hit}	beam residual tol.	0.30; 0.25 m
<i>Lifecycle validation (strict/loose)</i>		
τ_S	beam-support	0.75/0.25
τ_p	position	0.05/0.20 m
τ_ψ	yaw	2.0/10.0°
τ_U	posterior modes	1/2
–	support residual tol.	0.02 m
<i>Candidate / recovery</i>		
$d_{\text{cand}}, \psi_{\text{cand}}$	candidate motion	0.15 m, 30°
$d_{\text{rec}}, \psi_{\text{rec}}$	recovery motion	0.3 m, 100°
<i>Mapping</i>		
–	anchor grid	256^3
–	warm-start / online anchors	40k/20k
τ_w, τ_c, τ_s	conflict thresholds	0.5, 0.05, 0.05 m
λ_g, λ_e	gradient / Eikonal weights	0.018, 0.268
–	SDF loss / truncation	L_1 / 0.29 m

floorplan priors with observations.

These baselines include both explicit volumetric and implicit neural SDF representations, ensuring a comprehensive comparison across map paradigms.

Evaluation Metrics. *Alignment*: we report success, false-convergence, non-convergence, and failure rates, convergence frames, and pose error at verified convergence. A trial succeeds when the filter declares verified convergence and the estimated alignment satisfies $e_{xy} \leq 1.0$ m and $e_\psi \leq 30^\circ$; false convergence violates this threshold, non-convergence lacks verified convergence, and failure is their sum. For FP-AIM, the confirmed transform is computed from the candidate pose correspondence and the convergence frame marks successful candidate verification.

Mapping: we report SDF error, mesh accuracy, mesh completion, and Chamfer distance, with geometry extracted on a regular 256^3 grid. Accuracy and completion are bidirectional nearest-neighbor distances between sampled predicted and ground-truth surfaces (Chamfer is their average). SDF error is evaluated at 10^6 deterministically sampled valid query points for the neural methods; for nvblox we use its exported 256^3 ESDF grid, setting unknown cells to zero only for completion-sensitive evaluation.

Implementation Details. We emulate Crazyflie-class sparse sensing with six range beams at 20 Hz (four horizontal for alignment, all beams for mapping). All particle-filter variants use $N = 10,000$ particles and a 4.0 m sensing horizon; MCL/AMCL use $\sigma_{\text{hit}} = 0.30$ m and FP-AIM uses 0.25 m. The mapping module follows the LGSDf local-update rule and its defaults [10] ($\eta = 5$, $w_{\min} = 10^{-3}$). We summarize remaining hyperparameters, frozen on the validation split and fixed across all scenes in Table I. All experiments run on an

TABLE II
QUANTITATIVE RESULTS FOR GLOBAL ALIGNMENT. VALUES ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION ACROSS SEEDS. BEST RESULTS ARE SHOWN IN **BOLD**, AND SECOND-BEST RESULTS ARE UNDERLINED.

Method	Success (%) $\uparrow$	False Conv. (%) $\downarrow$	Non-conv. (%) $\downarrow$	Failure (%) $\downarrow$	Conv. Frames $\downarrow$	Trans. Err. (m) $\downarrow$	Rot. Err. (deg) $\downarrow$
AMCL	77.53 $\pm$ 0.86	15.56 $\pm$ 0.74	6.91 $\pm$ 1.13	22.47 $\pm$ 0.86	237.24 $\pm$ 10.30	0.0682 $\pm$ 0.0046	2.2027 $\pm$ 0.1458
MCL	62.22 $\pm$ 0.74	23.95 $\pm$ 2.60	13.83 $\pm$ 3.08	37.78 $\pm$ 0.74	139.37 $\pm$ 21.93	0.0662 $\pm$ 0.0055	2.2244 $\pm$ 0.0500
FP-AIM	85.19 $\pm$ 1.28	9.63 $\pm$ 0.74	5.19 $\pm$ 1.48	14.81 $\pm$ 1.28	<u>174.48 $\pm$ 22.85</u>	0.0682 $\pm$ 0.0067	1.7602 $\pm$ 0.1498

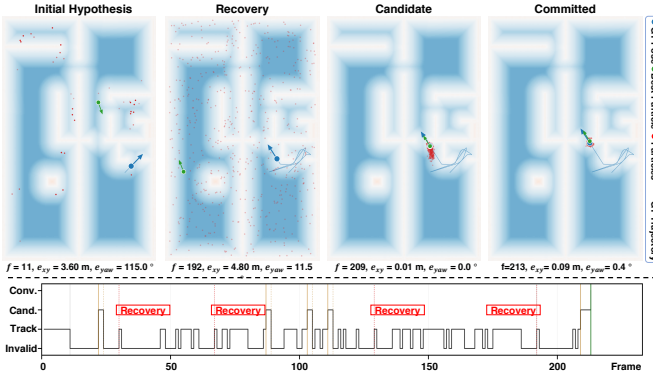


Fig. 3. Representative lifecycle of floorplan-aware global alignment. The filter initially supports an incorrect hypothesis, recovers from persistent tracking invalidity, and confirms the local-to-world transform only after candidate verification over accumulated motion. Pose errors are computed against the corresponding ground-truth poses.

AMD Ryzen Threadripper 7960X CPU, 128 GB RAM, and an NVIDIA RTX 4090 GPU.

B. Evaluation of Global Alignment

Table II reports global-alignment performance over 405 held-out test runs per method under sparse four-beam horizontal range observations. Proposed FP-AIM achieves the highest success rate, 85.19%, and the lowest failure rate, 14.81%, reducing both false convergence and non-convergence compared with AMCL and MCL, indicating more reliable alignment under extremely sparse range measurements.

MCL declares convergence earliest among successful trials, but it also shows the highest false-convergence and non-convergence rates. AMCL improves reliability at the cost of slower convergence. The FP-AIM alignment lifecycle instead treats sparse-beam particle-filter evidence as a provisional, discards unsupported candidates during verification, and globally reinitializes particles when tracking validity remains unsupported, requiring fewer convergence frames than AMCL while achieving the lowest overall failure rate.

Fig. 3 illustrates this behavior on Merom_0_int at h030, where low sensing height produces repeated tracking-invalid intervals due to floorplan-unmodeled objects. The lifecycle rejects unsupported candidates, recovers from persistent invalid tracking, and confirms the transform only after a later candidate remains supported through verification.

C. Evaluation of Floorplan-Aware Mapping

Table III evaluates mapping quality under sparse six-beam observations. FP-AIM_M isolates the mapping module with world-aligned poses, and FP-AIM_F uses the estimated alignment; FP-AIM_F stays within about 0.3 cm of FP-AIM_M on average (e.g., 3.12 vs. 2.79 cm SDF error), indicating limited degradation from the estimated alignment. FP-AIM attains the lowest errors across all metrics.

The baselines instead represent what is attainable from sparse observations alone, with two characteristic failure modes: the explicit mapper leaves large unobserved regions (nvblox, 37.94 vs. 10.93 cm completion), while the implicit baselines fill space but over-smooth (e.g., 12.65 vs. 2.79 cm SDF error). Both reflect a fundamental limit of sparse-only mapping under micro-quadrotor payload constraints—precisely the gap a floorplan prior is meant to close. Fig. 4 shows this qualitatively: in the red-boxed weakly-constrained regions the explicit baseline leaves holes while the implicit baselines over-smooth or spatially spread surfaces, whereas FP-AIM preserves room-scale structure while incorporating measurement-supported local deviations.

D. Ablation Study.

Floorplan-Aware Global Alignment. Table IV evaluates the alignment lifecycle. Removing recovery drops the success rate from 85.19% to 55.56% and increases both false convergence and non-convergence, indicating that recovery helps avoid persistent tracking of unsupported hypotheses. Removing candidate verification lowers non-convergence to 0.49% but increases false convergence from 9.63% to 16.79%, showing the trade-off between rapid convergence declaration and incorrect acceptance. These results support the combined use of recovery and candidate verification over accumulated motion.

Floorplan-Aware Implicit Mapping. We ablate the mapping module on Ihlen_0_int (Fig. 5). The full FP-AIM_M variant obtains 2.68 cm SDF error. The no-update variant (w/o MOU) acts as a *floorplan-only* reference, since without measurements every target reduces to d_i^{fp} : it reaches 7.48 cm, showing the prior captures room-scale layout but is insufficient by itself. Crucially, fusing measurements *without* conflict-aware reconciliation (w/o CHS) degrades to 16.29 cm—worse than the floorplan-only reference—so the gain comes from deciding when to trust the prior versus measurements, not from access to the prior. Removing FI causes only minor degradation (2.85 cm), indicating warm-starting mainly stabilizes training.

TABLE III

QUANTITATIVE 3D RECONSTRUCTION RESULTS AVERAGED OVER THREE SEEDS. AIM INDICATES FP-AIM. THE BEST AND SECOND-BEST RESULTS ARE SHOWN IN **BOLD** AND UNDERLINED, RESPECTIVELY.

Metrics	SDF Error [cm] ↓					Mesh Accuracy [cm] ↓					Mesh Completion [cm] ↓					Chamfer Distance [cm] ↓				
Methods	nvblox	iSDF	LGSDf	AIM_M	AIM_F	nvblox	iSDF	LGSDf	AIM_M	AIM_F	nvblox	iSDF	LGSDf	AIM_M	AIM_F	nvblox	iSDF	LGSDf	AIM_M	AIM_F
Rs_int	23.26	13.38	13.98	2.95	2.98	5.94	5.32	7.01	3.51	<u>3.61</u>	27.49	12.62	11.76	9.40	9.44	16.72	8.97	9.39	6.46	6.53
Merom_0_int	19.57	8.53	8.31	2.75	<u>3.47</u>	6.90	5.90	6.03	4.06	<u>5.24</u>	38.42	12.76	12.30	10.53	<u>10.87</u>	22.66	9.33	9.17	7.30	<u>8.06</u>
Ihlen_0_int	28.71	16.46	15.66	2.68	<u>2.92</u>	6.72	8.20	11.14	4.25	<u>4.43</u>	47.92	14.39	13.86	12.87	<u>12.95</u>	27.32	11.30	12.50	8.56	<u>8.69</u>
Average	23.85	12.79	12.65	2.79	<u>3.12</u>	6.52	6.47	8.06	3.94	<u>4.23</u>	37.94	13.26	12.64	10.93	<u>11.09</u>	22.23	9.87	10.35	7.44	<u>7.76</u>

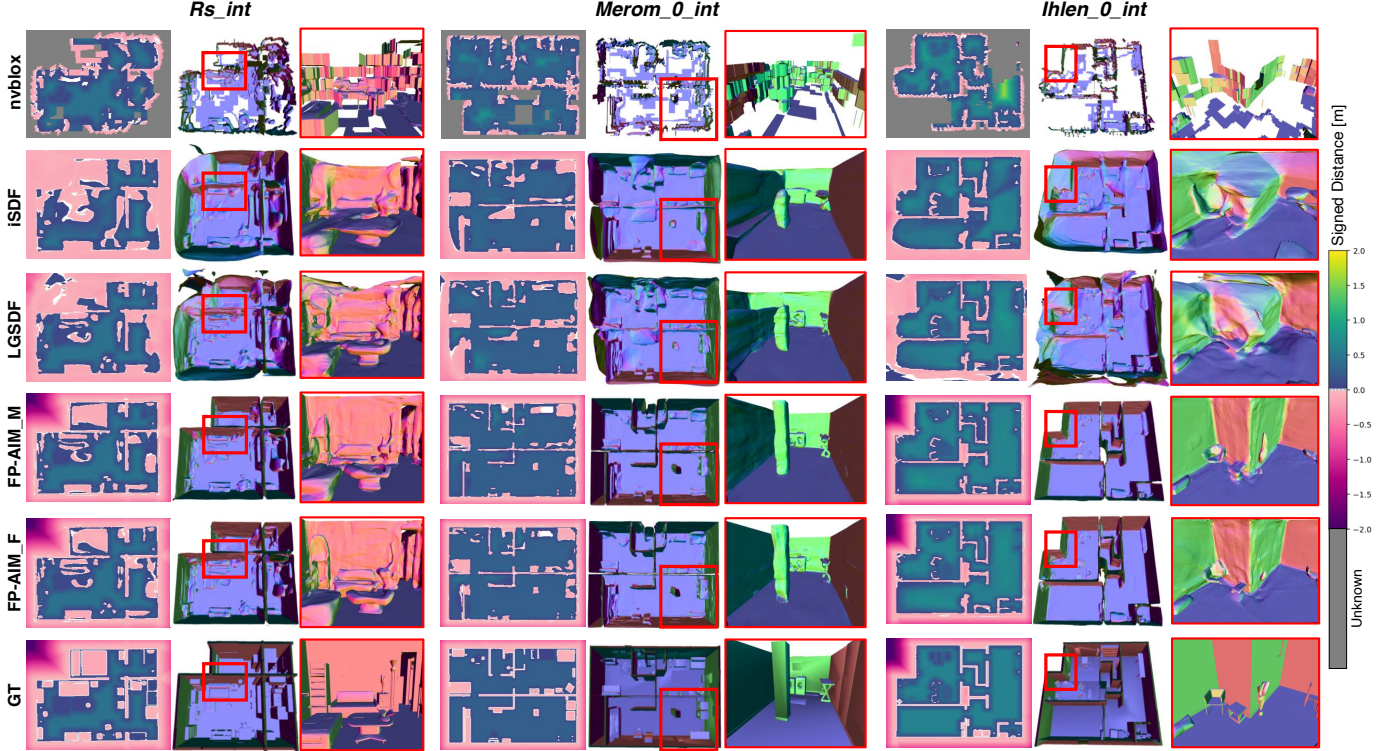


Fig. 4. Qualitative comparison of 3D reconstruction results on Rs_int, Merom_0_int, and Ihlen_0_int. The red boxes highlight weakly constrained regions near walls, ceilings, and nearby objects, where sparse observations can lead to over-smoothed or spatially spreading surfaces. FP-AIM preserves room-scale structural consistency while incorporating measurement-supported local deviations.

TABLE IV

ABLATION STUDY OF THE FP-AIM ALIGNMENT LIFECYCLE. VALUES ARE MEAN $\pm$ STANDARD DEVIATION ACROSS SEEDS.

Variant	Succ. ↑	False Conv. ↓	Non-conv. ↓	Failure ↓
Full	85.19 $\pm$ 1.28	9.63 $\pm$ 0.74	5.19 $\pm$ 1.48	14.81 $\pm$ 1.28
w/o Recovery	55.56 $\pm$ 1.96	21.73 $\pm$ 2.60	22.72 $\pm$ 1.54	44.44 $\pm$ 1.96
w/o Candidate verification	82.72 $\pm$ 0.43	16.79 $\pm$ 1.13	0.49 $\pm$ 0.86	17.28 $\pm$ 0.43

E. Real-world Experiment.

We conduct a qualitative full-pipeline real-world experiment in an indoor room scene, denoted M09, using a Crazyflie 2.1 Brushless platform equipped with a Multi-ranger deck and a Flow deck. We utilize the Flow-deck-aided EKF odometry as the local pose stream ${}^L T_B(t)$ and run the pipeline with sparse Multi-ranger measurements. For the LGSDf mapping baseline, we use the same sparse measurements with a manually world-aligned Flow-deck pose stream for a fair comparison.

As shown in Fig. 1, LGSDf produces more pronounced surface distortion and disconnected artifacts in the cluttered red-boxed region, whereas FP-AIM maintains a more continuous global layout and adapts local geometry around observed objects. In the yellow-boxed planar region, FP-AIM preserves a more coherent room-scale surface with fewer holes and less severe deformation. This qualitative result is consistent with the simulation trend under real sparse range sensing and onboard odometry.

F. Runtime Analysis.

We evaluate runtime on the same offboard GPU server used for all experiments. The FP-AIM alignment module processes one sparse range frame in 29.67 ms on average (33.70 Hz), exceeding the 20 Hz range stream. The FP-AIM mapping module performs one online refinement update in 55.70 ms (17.96 Hz), excluding the one-time floorplan warm-start and evaluation-only mesh extraction. The anchor buffer

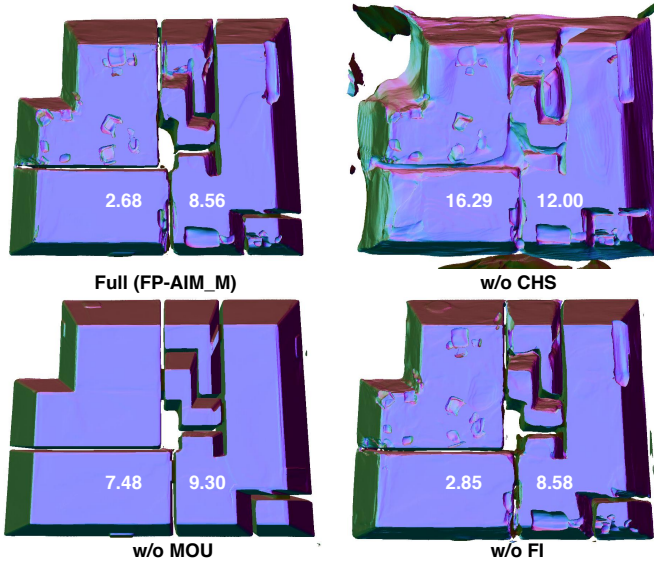


Fig. 5. Ablation study on the Ihlen_0_int scene, averaged over three random seeds. The two numbers in each image indicate SDF error and Chamfer distance in centimeters, respectively.

accumulates incoming supervision, so online refinement need not execute one optimizer update for every range frame.

V. CONCLUSION

This letter presented FP-AIM, a floorplan-aware framework for global alignment and implicit SDF mapping under sparse range sensing. The framework uses a 3D floorplan as a structural reference that connects local odometry, sparse range observations, and signed-distance reconstruction in a common floorplan-aligned frame. The alignment stage improves local-to-world transform estimation through sparse-beam likelihood evaluation, candidate verification, and recovery. For mapping, our results indicate that under sparse observation the floorplan prior is most effective when reconciled with measurements rather than trusted directly: conflict-aware supervision preserves room-scale structure while recovering measurement-supported local geometry, whereas relying on the prior alone or fusing measurements without conflict handling degrades reconstruction. Simulation-generated indoor sequences show improved alignment reliability and reconstruction quality, and a qualitative real-world deployment shows coherent reconstruction under sparse onboard range sensing.

One limitation is that the alignment stage estimates a single transform and is not refined afterward, so a small residual yaw error can manifest as a floorplan-measurement mismatch that slightly distorts room boundaries in the reconstruction. Future work will add post-alignment refinement to correct such residual errors, and will further address long-term local-odometry drift, larger floorplan-scene deviations, and onboard resource-aware optimization.

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